

Supplier Lead-Time Risk Prediction

Project Overview — DAP391m, Summer 2026

Group 8

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Abstract

Supplier lead-time variability poses significant stockout and procurement risks in retail operations. This study builds a predictive classification framework using supply chain logistics data, benchmarking four classifiers (Logistic Regression, Decision Tree, Random Forest, XGBoost) with SHAP explainability to identify high-risk suppliers.

Keywords: Supplier Risk Prediction, Delay Classification, Supply Chain Analytics

1. Project Objective

Supplier lead-time delays cause stockouts, margin erosion, and poor customer satisfaction in retail procurement. The goal of this project is to build a **predictive risk classification system** that flags high-risk shipments before they are delayed, giving procurement teams an early-warning signal they can act on.

The project answers three core research questions:

1. Which suppliers and routes carry the highest delay risk?
2. Which machine learning model best predicts shipment delays?
3. How can the predictions be explained and acted on by non-technical users?

Deliverables include a trained predictive model, an explainability report, a Power BI dashboard, and an interactive Streamlit web application.

2. Dataset

The analysis uses three publicly available CSV files from Kaggle, merged into a single modelling dataset of **704 rows and 23 features**. The source files contained no missing values in the columns retained for modelling; the inner joins described below further ensure that only complete records with matched supplier and regional metadata enter the final dataset.

Table 1: Data sources

File	Rows	What it contains
customer.csv	750	Supplier attributes, lead-time days, market segment
shipment.csv	728	Delivery status (target), customs time, freight cost
logistics_performance.csv	100	Carrier delay averages and warehouse utilisation by region

Data integration. `shipment.csv` serves as the main fact table, where each row corresponds to a single shipment and contains the target variable. `customer.csv` is joined as a supplier-attribute lookup on `supplier_id`, and `logistics_performance.csv` is joined as a regional-performance lookup on `region`. Inner joins are used so that only shipments with complete supplier and regional metadata are retained; this accounts for the reduction from the raw source rows to the final modelling dataset of 704 rows.

The target variable is whether a shipment was **Delayed** (16%) or **On-Time** (84%). Class imbalance is addressed through class weighting in model training and threshold tuning during evaluation.

3. Tools and Methods

Table 2: Tools and methods used

Area	Tools / Approach
Data processing	Python, SQL
Models compared	Logistic Regression, Decision Tree, Random Forest, XGBoost
Evaluation	PR-AUC and recall on the delayed class (primary, imbalance-aware); ROC-AUC, F1-score, and confusion matrix (secondary); Stratified 5-fold cross-validation
Explainability	SHAP (feature importance + per-shipment risk scores)
Visualisation	Plotly, Folium, Power BI supplier scorecard
Web application	Streamlit (input shipment → predict delay probability)
AI assistance	Claude (Anthropic), GitHub Copilot, Codex (OpenAI) — logged in AI Audit Log
Collaboration	GitHub, Google Drive, Overleaf

4. Project Description

Table 3: Project requirements and key points

Requirement	Key Points	Outcome	Week
Business Understanding & Analytic Approach — Identify problem — Define business impact	Lead-time variability directly causes stockouts; goal is to classify delay risks via classification models. Survival analysis concepts may be referenced as supporting interpretation but are not the primary modelling approach.	Clear problem scope and analytic direction	1
Research Questions (RBL) — Define hypotheses — Map questions to data	Which suppliers have highest delay risk? Which model best predicts delays? How can predictions be explained for non-technical users?	Research questions and evaluation criteria finalised	1
Data Collection, Understanding & Preparation — Collect from sources — Clean data	Kaggle supply chain dataset; features include actual vs. expected lead time and order volume; engineer binary delay target.	Clean dataset with delay target and engineered features	1
Data Analysis with SQL — Query database — Aggregate by region — YoY growth rate — Penetration index	Compute avg lead time per supplier, delay frequency, volume–delay correlation, and trends over time.	Supplier-level risk patterns identified	2
Data Analysis with Python — Train XGBoost classifier — Evaluate PR-AUC, recall, ROC-AUC, F1 — Confusion matrix analysis — Apply SHAP	XGBoost as primary classifier (4 models benchmarked); SHAP explains which features drive each supplier’s risk; evaluate primarily on PR-AUC and recall for the delayed class (imbalance-aware), with ROC-AUC, F1-score, and confusion matrix as secondary metrics.	Predictive model with explainable supplier risk scores	2
Data Visualisation — Boxplots by supplier — Lead-time histograms — Risk heatmap — Trend lines	Risk heatmap surfaces highest-risk supplier–route pairs; boxplots reveal lead-time spread per supplier; histograms and trend lines show distribution over time.	Visual evidence of lead-time variation and delay risk	3
Regression Analysis — Interpret logistic regression coefficients — Compute odds ratios — Identify significant delay drivers — Complement SHAP interpretation	Interpret the logistic regression trained in Week 2 by extracting odds ratios and significance of each feature; identify statistically explainable delay drivers to complement SHAP feature importance from XGBoost.	Statistical interpretation of delay drivers, complementing the XGBoost SHAP analysis	3
Data Analysis with Tool — Supplier scorecard — Risk alerts	Power BI supplier scorecard with automated risk alerts; drill-down by supplier, region, and carrier; supports procurement diversification decisions.	Interactive BI dashboard for supplier monitoring	4
Web Application — Input shipment parameters — Predict delay probability — SHAP explanation	Real-time delay probability from shipment inputs; SHAP waterfall explains each individual prediction; XGBoost model accessible to non-technical users.	Live risk prediction tool for procurement teams	4–5